POLITÉCNICO DE LEIRIA | ESCOLA SUPERIOR DE TECNOLOGIA E GESTÃO

Master Degree - Product Design Engineering
First Year / First Semester
2025/2026

Product Modelling and Simulation – FEM Work

Exercise 01 (6.0/20 points)

A steel plate is subjected to an axial load, as shown in Figure 1. Approximate the deflections and average stresses along the plate. The plate is 1/16 in thick and has a modulus of elasticity E = 29 x1 0^6 lb/in^2.

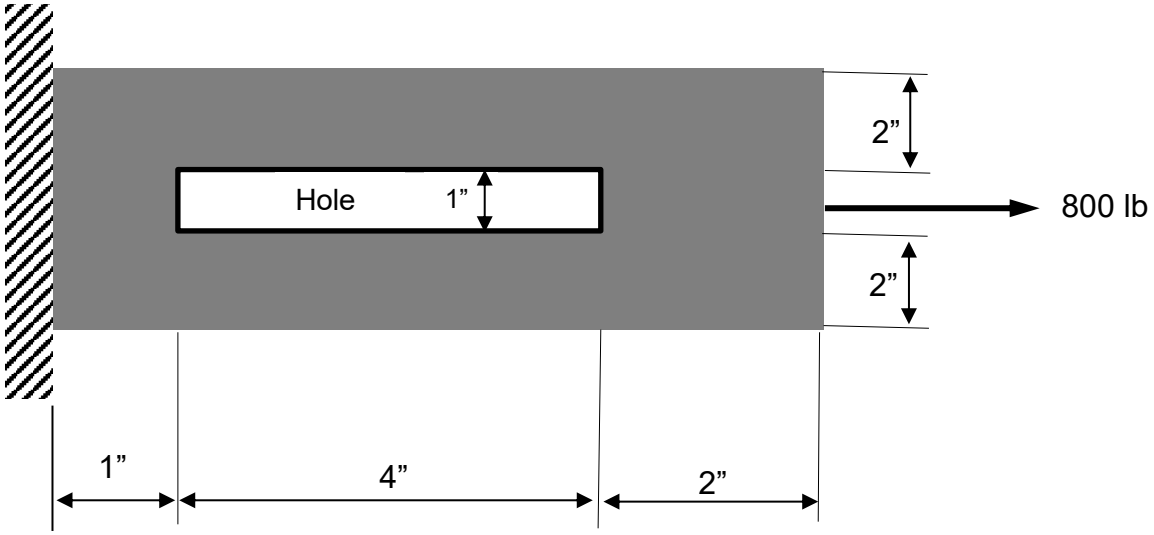


Figure 1

Exercise 02 (6.0/20 points)

Given the truss structure formed by four bars (Figure 2), the elastic Young’s modulus and cross-sectional area are E = 29.5×104 N/mm2, A = 100 mm2, for all four bars. The question is to find the nodal displacement, the elemental stress and the reaction force in the truss structure.

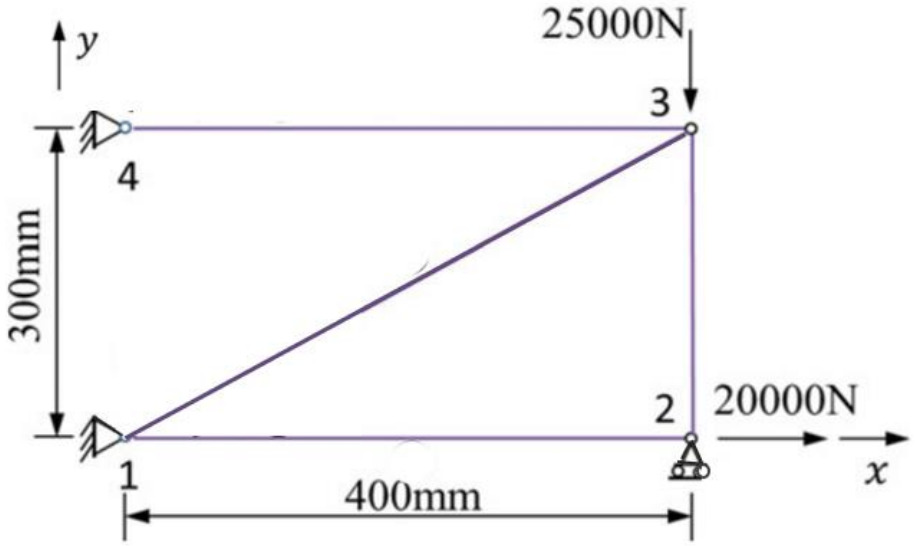


Figure 2

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Exercise 03 (8.0/20 points)

A three-dimensional truss is often called a space truss. For that, consider a simple space truss with six members joined together at their ends to form a tetrahedron, as shown in Figure 3.

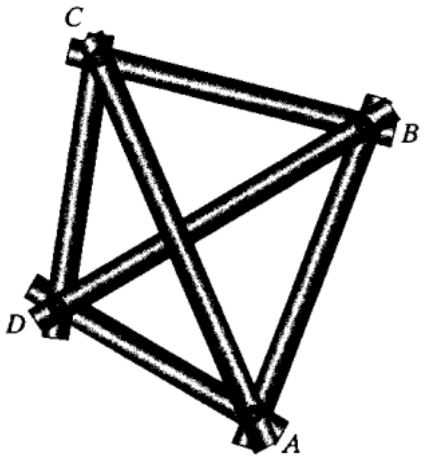


Figure 3: Simple space truss.

It is well known that members of a truss are generally considered to be two-force members. In the analysis of space trusses, it is assumed that the members are connected together by ball-and-socked joints with no bending moments at the joins. In addition, all loads must be applied at the joints and the weights of the members are usually negligible compared to the applied loads. However, if the wights of the members are to be considered, then half of the weight of each member is applied to the connecting joints.

In this context, the finite element formulation (FEM) of space trusses can be an extension of plane trusses. In a space truss, the global displacement of an element is represented by six unknowns, *Uix, Uiy, Uiz, Ujx, Ujy,* and *Ujz*, because each node (joint) can move in three directions, Moreover, the angles *Θx, Θy*, and *Θz* define the orientation of a member with respect to the global coordinate system. As the procedure for obtaining the finite element stiffness matrix for a space-truss is identical to the one we followed to derive the two-dimensional truss finite element stiffness, the question is to derive the finite element stiffness matrix $\mathbf{[K]^e}$ for a space-truss.

Observation: When solving the above problems, all relevant information must be described/presented.